OFFICIAL TECHNICAL DOCUMENT

Sterling Wholesale Insurance Portal

Enterprise Technical Architecture, Operational Workflow Manual & Complete API Directory

APPLICATION CORE
Next.js 14 (App Router) + TypeScript

PERSISTENCE & CLOUD TIER
MongoDB Atlas + Backblaze B2 (S3 SDK)

TARGET JURISDICTION
US Commercial Lines & Surplus Lines

PRIMARY MODULES
Builders Risk, GL, Inland Marine, Habitational

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November 2026

REPOSITORY TARGET
eliteoperationglobal/sterling (main)

Sterling Wholesale Insurance Services LLC

Confidential & Proprietary Developer Documentation

1. Executive Summary & Core Mission

The **Sterling Wholesale Insurance Portal** is a specialized enterprise brokerage management solution engineered to solve the complex underwriting, statutory compliance, rating, and binding challenges inherent to US commercial surplus lines insurance.

> **SYSTEM MISSION & PURPOSE**
>
> To eliminate latency in the wholesale placement pipeline by providing an automated, multi-party bridge connecting retail insurance producers, wholesale underwriters, surplus lines stamping offices (e.g. California SLA, SLTX), premium finance companies, and insurance carriers.

High-Level System Capabilities

🏢 Retail Agency Portal

Provides retail insurance brokers with a digital command center to submit commercial risks, track underwriting status, customize retail broker fees, dispatch client e-sign envelopes, and request immediate policy binding.

⚙️ Wholesale Underwriter Console

Enables wholesale underwriters to manage risks across all retail agencies, evaluate loss runs, enter carrier terms, automate state surplus lines tax and stamping calculations, issue PDF proposals, and approve binds.

🔗 Public Prospect Intake

Retail agents generate tokenized, secure single-use web links allowing prospective clients to fill out complete commercial applications and upload documents without requiring an agency login.

💳 Integrated Premium Financing

Real-time EMI loan calculator supporting 10% to 25% down payments and 9-to-11 monthly installments, automatically creating formal Premium Finance Agreements (PFA).

Core Architecture Topology

LAYER	COMPONENT	IMPLEMENTATION DETAILS
Frontend & API	Next.js 14.2 (App Router)	React Server Components (RSC), Tailwind CSS, Lucide Icons, Client Hydration via SWR.
Database Tier	MongoDB Atlas & Mongoose 8.3	Document database storing polymorphic submissions, multi-policy quotes, agencies, and audit logs.
Identity & Auth	NextAuth.js v4 + JWT + 2FA	Credential-based JWT session strategy with optional 6-digit email OTP 2FA and trusted device fingerprinting.
Cloud Storage	Backblaze B2 (AWS S3 SDK v3)	Presigned uploads and downloads for policy binders, invoices, signed accords, and uploaded loss runs.
PDF Generation	Puppeteer & Sparticuz Chromium	Server-side HTML-to-PDF rendering for ACORD 125/126, proposals, and invoices.

LAYER	COMPONENT	IMPLEMENTATION DETAILS
Notifications	Resend SDK & Nodemailer	Automated transactional dispatch for quote delivery, OTP login codes, and bind confirmations.

2. Supported Commercial Programs & Data Models

The platform accommodates diverse commercial P&C underwriting programs, each with tailored data models, validation schemas, and ACORD mappings:

1. Builders Risk

Specialized ground-up residential and commercial construction, renovations, and remodeling risks.

CATEGORY	CAPTURED FIELDS & UNDERWRITING LIMITS
Core Limits	Limit per Occurrence, Completed Value, Total Insurable Value, Existing Structure, Valuation (ACV vs Replacement Cost), Coinsurance (80%, 90%, 100%).
Sublimits	Water Damage, Soft Costs, Delay in Completion, Property in Transit, Temporary Offsite Storage, Debris Removal, Pollutant Clean-up, Scaffolding, Fire Dept Service Charge.
Deductibles & Values	All-Perils Deductible, Fire Deductible, Building Value.

2. Contractors Equipment (Inland Marine)

Covers mobile equipment, heavy machinery, tools, and accessories utilized by trade contractors.

- **Scheduled Equipment**: Specific items itemized by year, make, model, serial number, and individual insurable limit.
- **Unscheduled Tools & Employee Tools**: Blanket coverage limit with maximum per-item caps ($2,500 / $5,000).
- **Rented / Leased Equipment**: Short-term and long-term equipment rental coverage with deductible options.

3. Manufacturers

Commercial coverage for light industrial, food processing, metal fabrication, and goods manufacturing.

- **Commercial General Liability (CGL)**: Occurrence & Aggregate limits ($1M / $2M), Products & Completed Operations.
- **Commercial Property**: Building value, Business Personal Property (BPP), and Equipment Breakdown.
- **Specialized Add-ons**: Independent radio selection for *Professional Liability* and *Pollution Liability*.

4. Apartment Buildings & Habitational

Multi-family residential risks, apartment complexes, mixed-use, and commercial residential properties.

- **Habitational Questionnaire**: Roof age, wiring updates, plumbing pipe material, hardwired smoke detectors, tenant safety.
- **Limits & Endorsements**: Building coverage, Business Income / Extra Expense, Building Ordinance & Law.

3. End-to-End Operational Lifecycle

A commercial submission transitions through distinct statutory and operational states from initial intake to final policy issuance:

1. **Risk Intake (Draft & Submit)**
   The retail broker creates a submission in the agency portal or a prospective client submits via a tokenized link (`/apply/[token]`). The intake engine validates required attachments (Loss Runs, ACORD forms) and creates a `Submission` document with status `SUBMITTED`.

2. **Underwriting & Multi-Carrier Quote Entry**
   Wholesale underwriters inspect the risk, market to carriers (e.g. Kinsale, Richmond National, Lloyd's), and enter formal quote terms into `/admin/submissions/[id]/quote`. The system automatically computes state surplus lines tax and stamping fees.

3. **Proposal Publication & Broker Markup**
   The underwriter posts the quote to the retail broker. The retail agent reviews terms, inputs their retail broker fee, and downloads the pixel-perfect Proposal PDF. The submission status updates to `QUOTED` / `ACCEPTED_BY_AGENCY`.

4. **E-Signature & Payment Capture**
   The system generates the complete application packet, ACORD 125/126 forms, and Payment Authorization / Premium Finance Agreement. Digital e-signatures are captured from the insured and broker. Payment is authorized via Credit Card, ACH, or Financed down payment.

5. **Bind Request & Policy Issuance**
   The agent clicks 'Request Bind'. The underwriter verifies subjectivities and payment confirmation, approves the order, uploads the final policy binder, and issues the policy. The status becomes `BOUND`.

6. **Post-Bind Administration & Policy Audits**
   The platform supports mid-term policy endorsements, cancellations, and premium audits via `/admin/audit` and `/admin/cancel-requests`.

State Surplus Lines Tax & Stamping Matrix

STATE JURISDICTION	SURPLUS LINES TAX RATE	STAMPING OFFICE FEE	STATUTORY NOTES
California (CA)	3.00%	0.250% (SLA CA)	Calculated on Gross Premium + Carrier Fees + Wholesale Policy Fee.
Texas (TX)	4.85%	0.040% (SLTX)	Surplus Lines Stamping Office of Texas reporting.
Florida (FL)	4.94%	0.060% (FSLSO)	Florida Surplus Lines Service Office filing requirements.
All Other States	Dynamic Lookup	State Stamping Fee	Engine references `src/lib/data/stateTaxRates.ts` by property risk state.

4. Core RESTful API Directory

The platform exposes over 100 modular route handlers structured across clean administrative, agency, and public domains:

Administrative Endpoints (`/api/admin/*`)

ENDPOINT	HTTP	FILE ROUTE	OPERATIONAL DESCRIPTION
`/api/admin/submissions`	`GET, POST`	`admin/submissions/route.ts`	Query, filter, and paginate wholesale submissions across all agencies.
`/api/admin/submissions/[id]`	`GET, PUT`	`admin/submissions/[id]/route.ts`	Inspect submission details and update risk specifications.
`/api/admin/quotes/[id]/enter`	`POST`	`admin/quotes/[id]/enter/route.ts`	Persist carrier quote limits, deductibles, premiums, and calculate taxes.
`/api/admin/quotes/[id]/post`	`POST`	`admin/quotes/[id]/post/route.ts`	Publish completed carrier quote to the retail broker.
`/api/admin/bind-requests/approve`	`POST`	`admin/bind-requests/approve/route.ts`	Approve retail bind order, generate bound policy, and dispatch alerts.
`/api/admin/audits`	`GET, POST`	`admin/audits/route.ts`	Manage policy premium audits and mid-term adjustment invoices.

ENDPOINT	HTTP	FILE ROUTE	OPERATIONAL DESCRIPTION
`/api/admin/submissions/[id]/proposal`	`GET`	`admin/submissions/[id]/proposal/route.ts`	Stream real-time rendered Puppeteer proposal PDF for underwriter.

Agency Endpoints (`/api/agency/*`)

ENDPOINT	HTTP	FILE ROUTE	OPERATIONAL DESCRIPTION
`/api/agency/submissions`	`GET, POST`	`agency/submissions/route.ts`	Retrieve submissions scoped to the authenticated agency context.
`/api/agency/applications/submit`	`POST`	`agency/applications/submit/route.ts`	Direct risk application intake with multi-file attachment parsing.
`/api/agency/links/generate`	`POST`	`agency/links/generate/route.ts`	Create cryptographically secure, tokenized intake links for clients.
`/api/agency/quotes/[id]`	`GET`	`agency/quotes/[id]/route.ts`	Fetch received carrier quote terms and configure broker fee markup.
`/api/agency/submissions/[id]/request-bind`	`POST`	`agency/.../request-bind/route.ts`	Submit bind order to wholesale underwriting with payment authorization.

Public & Utility Endpoints (`/api/*`)

ENDPOINT	HTTP	FILE ROUTE	OPERATIONAL DESCRIPTION
`/api/auth/[...nextauth]`	`GET, POST`	`auth/[...nextauth]/route.ts`	NextAuth credential verification, JWT generation, and 2FA OTP handler.

ENDPOINT	HTTP	FILE ROUTE	OPERATIONAL DESCRIPTION
`/api/public/links/verify`	`GET`	`public/links/verify/route.ts`	Validate single-use dynamic link token for prospective client intake.
`/api/public/applications/submit`	`POST`	`public/.../submit/route.ts`	Unauthenticated client submission intake via verified token.
`/api/finance/quote`	`POST`	`finance/quote/route.ts`	Calculate EMI loan installments, down payments, APR, and PFA terms.
`/api/tax/calculate`	`POST`	`tax/calculate/route.ts`	Calculate statutory state surplus lines tax and stamping fees.

5. Installation, Deployment & System Maintenance

Local Development Setup

```
# 1. Clone repository & install dependencies
git clone https://github.com/eliteoperationglobal/sterling.git
cd sterling
npm install

# 2. Configure environment file
cp .env.example .env.local
# (Populate MONGODB_URI, NEXTAUTH_SECRET, JWT_SECRET, etc.)

# 3. Seed database with core agencies, carriers, forms & admin
npm run seed

# 4. Start local development server
npm run dev
# Running at: http://localhost:3000
```

Deployment Topologies

▲ Vercel Serverless

Connect repository directly to Vercel. Puppeteer uses `@sparticuz/chromium` for serverless PDF rendering. Configure environment variables in the project dashboard.

🐳 Docker & Docker Compose

Execute `docker-compose up -d --build`. Utilizes multi-stage standalone Node.js container with integrated dependencies on port 3000.

Pre-Deployment Verification Script

An automated diagnostic script is provided in the repository to verify environment variables, git sync, and import integrity prior to deploying to staging or production:

```
powershell -ExecutionPolicy Bypass -File .\check-deployment-ready.ps1
```

> **DOCUMENT VERIFICATION & MAINTENANCE**
>
> This document represents the official architectural baseline for **Sterling Wholesale Insurance Services**. For questions or system upgrades, refer to the active repository at github.com/eliteoperationglobal/sterling.